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TryHackMe Writeup

Date Completed: June 11, 2026

Room Title: AI/ML Security Threats

Link: <https://tryhackme.com/room/aimlsecuritythreats>

Focus Area: How is AI used in Cybersecurity

Executive Summary

Completed the **AI/ML Security Threats** room on TryHackMe. This room provided an introduction to artificial intelligence, machine learning, deep learning, and their growing impact on the cybersecurity industry. Throughout the room, I explored the foundational concepts behind modern AI systems, including neural networks, training methodologies, large language models (LLMs), and transformer architectures. The training highlighted how advancements in AI have enabled systems to process and analyze large volumes of information efficiently, making them valuable tools for both cybersecurity professionals and threat actors.

The room also examined several security risks associated with AI technologies, including prompt injection, data poisoning, model theft, privacy leakage, and model drift. These threats demonstrate how AI systems can be manipulated or exploited if proper safeguards are not implemented. Additionally, the room covered how attackers can leverage AI to enhance phishing campaigns, generate malware, and create convincing deepfakes. Finally, the training emphasized the defensive applications of AI in cybersecurity, such as threat detection, log analysis, threat hunting, incident investigation, and automated reporting. Overall, this room reinforced the importance of understanding both the benefits and risks of AI, as the technology will continue to play a significant role in both cyber offense and defense.

Learning Objective

- Understand AI, ML, and their impact on the cyber security industry.

- Understand Deep Learning (DL) and neural networks, and how they have made the applications of AI, we see today, possible.
 - Understand how adversaries use AI to enhance existing attacks and take advantage of AI Model vulnerabilities.
 - Understand the key role AI will play in defending against AI.
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Walkthrough

Task 1: Introduction

Task 2: This task discusses machine learning, deep learning, and neural networks. Specifically, it talks briefly about the different training methods for machine learning and what information is needed to train the different models.

Task 3: This task discusses large language models (LLMs) and how they are trained and used to make AI models. This is a huge part of what modern AI uses to write code, give good feedback, and write papers. Transformer neural networks revolutionized LLMs by allowing the AI to process in parallel rather than sequentially. User feedback allows the AI to correct weak or completely wrong responses to the user's prompts.

Task 4: This task discusses how some of the impacts that AI has made on the cybersecurity industry. Specifically, AI introduces new threats such as:

- Prompt Injection: Using a prompt to override the AI and have it disclose information or access that it should not. This can produce either sensitive information, hateful content, or other unintended responses.
- Data Poisoning: Manipulating the data that the AI is trained on and references. This can result in a different output or stop security mechanisms that the AI is responsible for.
- Model Theft: Using data from an AI or API and training an AI model on that data to replicate the behavior of the original. Essentially, making your own clone of the AI model.
- Privacy Leakage: An AI model inadvertently reveals information that it is not supposed to. Private data is made publicly available.

- **Model Drift:** The AI stops working as intended over time. This requires retraining the AI and making sure it is working correctly. AI normally deteriorates or shifts slowly over time without intervention.

This task also discusses how AI can be used to run attacks on systems, create malware, deepfakes, and phishing.

Task 5: This task discusses how AI can drastically reduce recovery costs and time. Specifically, cybersecurity specialists can use AI's ability to analyze, predict, investigate, and summarize to reduce the grunt work that professionals have to go through in order to find security breaches and vulnerabilities in their systems.

Task 6: This task discusses how their chat AI can help perform various forms of analysis, such as log analysis, threat hunting, or content generation.

Task 7: Conclusion

Screenshots showing room completion

AI Security > AI Fundamentals > AI/ML Security Threats



AI/ML Security Threats

Learn AI basics, key terms, and how it's used by both attackers and defenders.

60 min

51,620

Share your achievement

Show Split View

Start AttackBox

Save Room

684 Recommend

Connectivity

Options

Room completed (100%)

Task 1 Introduction

Task 2 The Building Blocks of AI

Task 3 LLMs

Task 4 AI Security Threats

Task 5 Defensive AI

Task 6 Practical

Task 7 Conclusion

How likely are you to recommend this room to others?

1

2

3

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5

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7

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9

10

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120915

Badges

17

Streak

1

Completed rooms

90

Completed rooms

Certificates

Skills matrix

Badges

Created rooms

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2025 ReCapMe



Introduction to SIEM

Learn the fundamentals of SIEM and explore its features and functionality.

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Logs Fundamentals

Learn what logs are and how to analyze them for effective investigation.

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Firewall Fundamentals

Learn about firewalls and get hands-on with Windows and Linux built-in firewalls.

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IDS Fundamentals

Learn the fundamentals of IDS, along with the experience of working with Snort.

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Vulnerability Scanner Overview

Learn about vulnerability scanners and how they work in a practical scenario.

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CyberChef: The Basics

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CAPA: The Basics

Learn to use CAPA to identify malicious capabilities.

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FlareVM: Arsenal of Tools

Learn the arsenal of investigative tool in FlareVM.

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REMnux: Getting Started

Learn how you can use the tools inside the REMnux VM.

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AI/ML Security Threats

Learn AI basics, key terms, and how it's used by both attackers and defenders.

Easy

